Software Requirements Specification (SRS v2.0)

RestaurantApp — Food Ordering & Table Reservation System

> **Project:** RestaurantApp • **Version:** 2.0 (Final Submission — Firebase Only)
> **Target Platforms:** iOS & Android (Expo Router / React Native)
> **Backend:** Node.js / Express REST API (Port 5000)
> **Database Engine:** Google Firebase Firestore

1. Scope Changes & Architecture

All MongoDB and Mongoose dependencies have been removed. The backend runs solely on **Firebase Firestore** using the **Firebase Admin SDK**. The backend port has been updated to **Port 5000**.

2. Glossary of Terms

1. **RBAC (Role-Based Access Control):** Security architecture restricting managerial operations to authorized accounts.
2. **JWT (JSON Web Token):** Signed cryptographic token carrying user claims with a 1-day expiration.
3. **Bcrypt:** Password hashing function applied with 12 salt rounds before storage in Firestore.
4. **Idempotency:** Safe repeated script execution without duplicating Firestore records.
5. **NoSQL Document Collection:** Schema structure grouping documents into

collections (`users` , `menuItems` , `tables` , `orders` , `reservations`).
6. **Dining Slot Window:** A 90-minute time window used to detect overlapping reservation conflicts on the same table.

3. Verification & Automated Test Results

The backend features an automated Jest test suite (`server/tests/api.test.js`) verifying all 32 endpoints and business rules:

> **Automated Test Status:** PASS (32/32 Tests Passing) • **Frontend Typecheck:** 0 errors (`npx tsc --noEmit`)

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